**EXPERIMENT – 2**

# Encryption – Crypto 101 using TryHackMe Platform

## Aim

To understand the fundamentals of **cryptography** by performing hands-on exercises in the **Crypto 101** room on the **TryHackMe** platform, including basic encryption, encoding, hashing, and cryptographic concepts.

## Tools / Platform Required

- Web browser (Chrome / Firefox)
- Internet connection
- TryHackMe account (free)
- Built-in TryHackMe terminal (AttackBox) or local Linux machine

## Theory (Brief)

Cryptography is the practice of securing information by converting plain text into unreadable format (cipher text) using encryption techniques.
**Crypto 101** introduces basic concepts such as:

- Encoding vs Encryption
- Symmetric and Asymmetric encryption
- Hashing
- Common ciphers (Caesar, Base64, etc.)

## Steps to Perform the Experiment

### Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

### Step 2: Open Crypto 101 Room

1. In the search bar, type **Crypto 101**
2. Click on the room named **Crypto 101**
3. Click **Join Room**

### Step 3: Start the Machine (if required)

1. Click **Start Machine / Start AttackBox**
2. Wait for the virtual environment to load.

### Step 4: Perform Cryptography Tasks

Complete the tasks provided in the room, such as:

# 4.1 Encoding & Decoding

## (a) Base64 Encoding / Decoding

**Purpose:**
Base64 is an **encoding technique** used to represent binary data in readable ASCII format.

## How to Solve

**To Decode Base64**

1. Copy the given Base64 string from the question.

2. Open the TryHackMe terminal.

3. Use the command:

**echo "Q3J5cHRvZ3JhcGh5" | base64 -d**

4. The output will be the decoded plain text.

5. Enter the decoded text into the answer box.

To Encode Base64
**echo "cryptography" | base64**

**Expected Output Example: Cryptography**

### (b) Hexadecimal Conversion

**Purpose:**
Hexadecimal is a base-16 encoding system commonly used in computing**.**

### How to Solve

Hex to Text
**echo 48656c6c6f | xxd -r -p**

Text to Hex
**echo "Hello" | xxd -p**

**Expected Output: Hello**

## 4.2 Classical Ciphers

### (a) Caesar Cipher

Purpose:
Caesar cipher shifts letters by a fixed number.

### How to Solve

1. Identify the shift value (given or by trial).

2. Use online Caesar decoder or terminal tool.

**Using terminal**

```
echo "KHOOR" | tr 'A-Z' 'X-ZA-W'
```

(Shift = 3)
Expected Output: HELLO

## (b) Shift-Based Encryption

**Purpose:**
Similar to Caesar cipher but may use different shift values.

## How to Solve

1. Try different shift values until meaningful text appears.

2. Online tools can be used if allowed.

Example:

- Shift 1 → incorrect

- Shift 3 → readable text → correct answer

# 4.3 Hashing

## **(a)** Understanding Hash Algorithms

| Hash Type | Length |
|---|---|
| MD5 | 32 characters |
| SHA-1 | 40 characters |
| SHA-256 | 64 characters |

## (b) Identifying Hashes

How to Solve

1. Count the number of characters in the given hash.
2. Match it with the table above.

Example:

5d41402abc4b2a76b9719d911017c592

→ 32 characters → MD5

**Generating Hashes**

```
echo -n "hello" | md5sum
echo -n "hello" | sha1sum
echo -n "hello" | sha256sum
```

**4.4 Encryption Concepts**

**(a) Encryption vs Hashing**

| Encryption | Hashing |
|---|---|
| Reversible | Irreversible |
| Uses keys | No keys |
| Used for secure communication | Used for password storage |

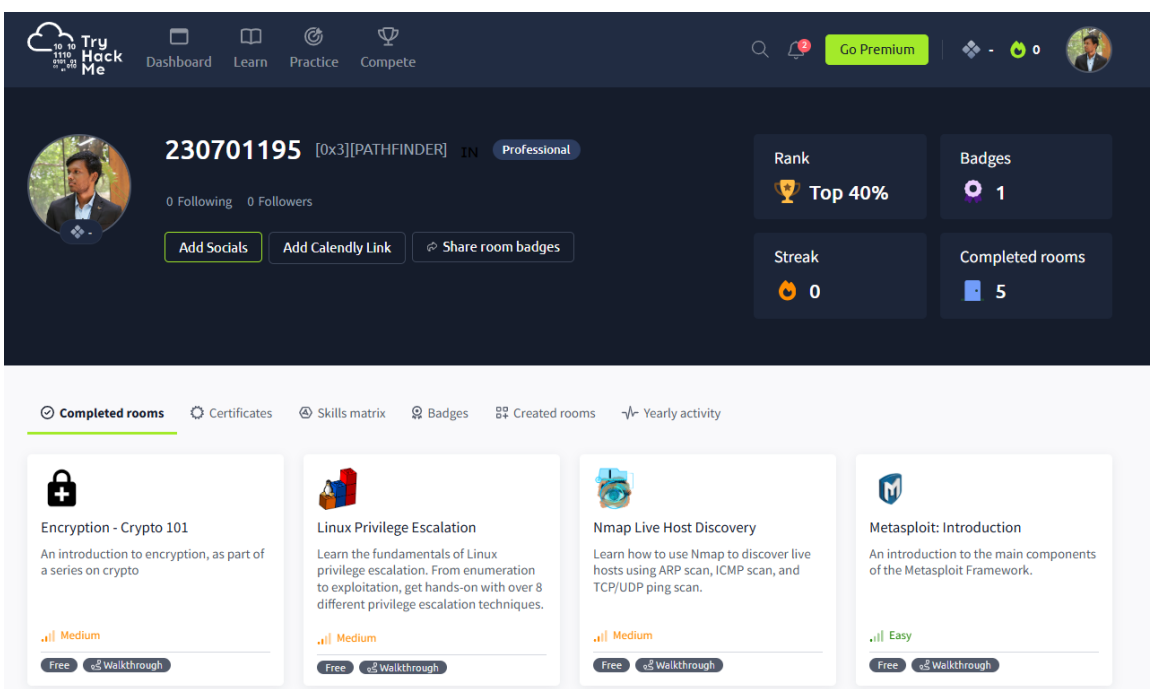
**(b) Symmetric vs Asymmetric Encryption**

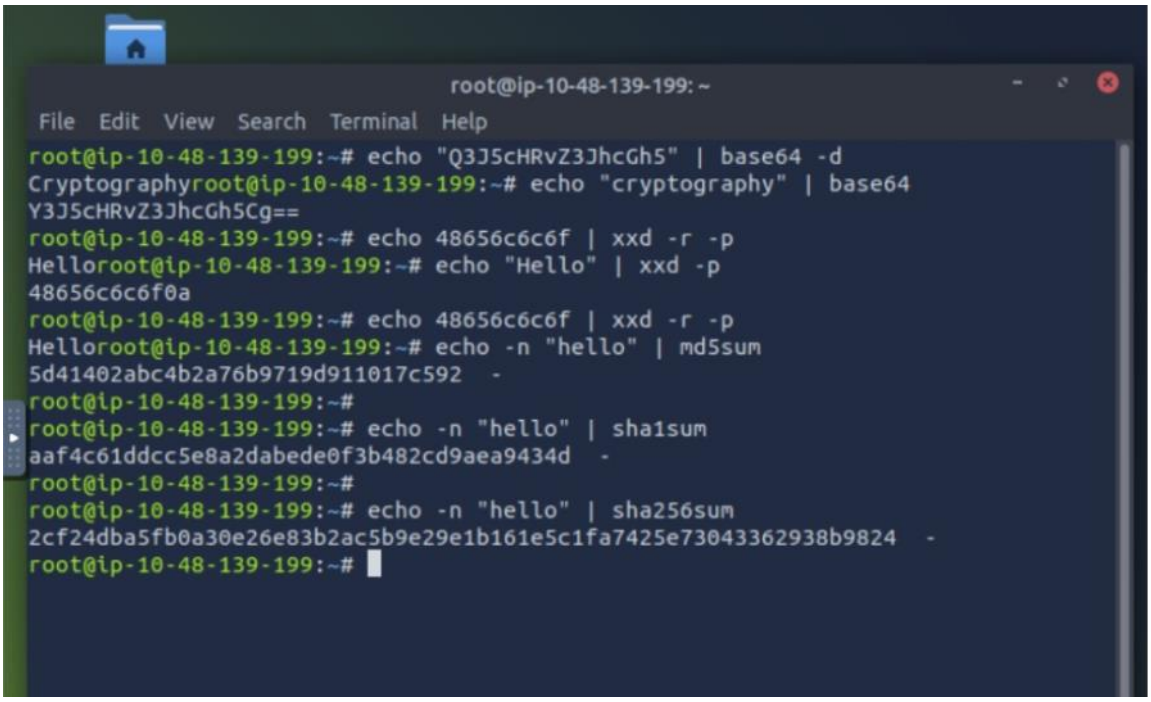
| Symmetric | Asymmetric |
|---|---|
| One key | Two keys |
| Fast | Slower |
| Example: AES | Example: RSA |

## Step 5: Verify Completion

1. Ensure all tasks are marked **Completed**
2. Observe the success message at the end of the room

# Output






# Result

Thus, the **Crypto 101** experiment was successfully completed on the **TryHackMe platform**.
The experiment helped in understanding **basic cryptographic concepts**, encryption techniques, encoding methods, and hashing algorithms through hands-on practice.